

3.4 SHORT RUN OR LONG RUN?

Multiple Choice Questions

1994/CE/II/22

A producer increases all his input to double the output. He finds that the total production cost increases by 80%. This can be explained by

- A. the law of supply.
- B. the law of diminishing returns.
- C. economies of scale.
- D. diseconomies of scale.

1996/CE/II/24

No. of machines	No. of workers	Total output
5	7	100
5	8	120
5	9	150
5	10	200

Refer to the above table. Which of the following is a possible reason for the increase in total output?

- A. division of labour
- B. the law of diminishing returns
- C. the law of comparative advantage
- D. economies of scale

1997/CE/II/15

During the busy season before Christmas, a factory increases the number of workers to increase output. The result is shown below:

No. of workers	Total product (Units)
21	210
22	242
23	276
24	300
25	310

The change in output can be explained by

- A. the law of diminishing returns.
- B. diseconomies of scale.
- C. (out of syllabus)
- D. (out of syllabus)

2000/CE/II/16

Number of machines	Number of workers	Total output (units)
3	3	90
3	4	160
3	5	250
3	6	360

The above data refer to a _____ situation in which _____.

- A. short run the law of diminishing returns applies
- B. short run the law of diminishing returns does not apply
- C. long run economies of scale exist
- D. long run diseconomies of scale exist

2001/CE/II/17

The table below shows the production of Good X in a firm. Only two inputs are required in the production process.

Capital (Units)	Labour (Units)	Total Output (units)	Total Cost (\$)
1	1	20	100
2	2	35	200
3	3	45	300
4	4	50	400

From the above data,

- A. the law of diminishing marginal returns applies to the firm.
- B. the firm enjoys economies of scale.
- C. diseconomies of scale exist for the firm.
- D. the firm produces at its optimal scale of production.

2005/CE/II/19

The table below shows the production of Good X in a firm. Only two inputs are required in the production process.

Capital (Units)	Number of Workers	Average working hours of workers per day	Total Output
2	10	10	5 000
2	10	9	4 900
2	10	8	4 700
2	10	7	4 400

Due to a fall in market demand, the firm reduces the output by cutting the working hours of the workers from 10 hours a day to 8 hours a day.

Based on the above information, we can conclude that

- A. the supply of Good X by the firm decreases.
- B. the amount of labour employed by the firm remains constant.
- C. the law of diminishing marginal returns applies to the firm.
- D. the firm enjoys economies of scale.

2006/CE/II/19

A producer increases all his inputs to double the output. The following table shows the change in average cost after a change in output.

Total Output (Units)	Average Cost (\$)
10 000	6
20 000	7

Based on the above information, we can conclude that

- A. the firm enjoys economies of scale.
- B. the firm suffers from diseconomies of scale.
- C. the law of diminishing marginal returns applies to the firm.
- D. the firm produces at the optimal scale of production.

2007/CE/II/19

The input-output relationship of Firm A is as follows :

Capital (units)	Labour (units)	Marginal output (units)
1	10	10
1	11	9
1	12	8
1	13	7

From the above table, we can conclude that, as the labour input increases

- A. total output is increasing.
- B. average output is increasing.
- C. firm A is experiencing diseconomies of scale.
- D. the average cost of production is decreasing.

2008/CE/II/18

When a firm expands production by only employing more labour,

- A. the wage rate of its workers will increase.
- B. the average productivity of its workers will increase.
- C. the firm will enjoy economies of scale.
- D. the marginal product of its workers will eventually decrease.

2009/CE/II/19

Refer to the following information about the inputs, outputs and average cost of a firm. The only inputs required for production are capital and labour.

Capital (units)	Labour (units)	Total output (units)	Average cost (\$)
500	5	3 000	3
1 000	10	4 500	4
1 500	15	5 500	5

Within the above range of output, we can conclude that

- A. the firm is producing at the optimal scale of production.
- B. the law of diminishing marginal returns applies to the firm.
- C. the firm does not have a comparative advantage in the production of this good.
- D. the firm suffers from diseconomies of scale.

The following shows the cost-output relationship of a firm.

Output (units)	0	1	2	3	4	5
Total cost (\$)	10	28	42	52	60	72

Based on the above table, we can conclude that

- A. the firm enjoys the benefits of economies of scale.
- B. the firm's optimal scale of production is 5 units of outputs.
- C. the average variable cost is increasing.
- D. the total fixed cost is \$10.

2013/DSE/I/04

Machines (unit)	Labour (unit)	Average Product (unit)
2	3	30
2	4	40
2	5	50
2	6	55

The above table shows a _____ situation in which _____.

- A. short run the law of diminishing returns does not apply
- B. short run the law of diminishing returns applies
- C. long run economies of scale exists
- D. long run diseconomies of scale exists

2019/DSE/I/05

The following table shows the production data of a firm. The only inputs required are capital and labour.

Capital (units)	Labour (units)	Total product (units)	Total cost of production (\$)
2	2	10	100
3	3	30	270
4	4	45	360
5	5	55	385

Based on the above table, we can conclude that

- A. the law of diminishing returns applies to the firm.
- B. the total fixed cost is \$100.
- C. the optimal scale is 45 units.
- D. the firm enjoys economies of scale.

2020/DSE/I/8

The following table shows the production plan of a firm which employs two factors only, namely labour and machinery.

Machinery (units)	Labour (units)	Average product of labour (units)
4	1	20
4	2	21
4	3	22
4	4	X

If the above data illustrate the law of diminishing marginal returns, the value of X _____.

- A. can be 23
- B. can be smaller than 22
- C. must be larger than 22
- D. must be smaller than 22

Short & Structured Questions

1993/CE/I/1(b)

Yummy Yummy is a fast food shop. With more customers, 'Yummy Yummy' had to employ more workers to increase the output. However, with its existing facilities and equipment, the shop could not further increase the output by employing more workers when the number of workers had reached a certain level.

Name and state the economic law which explains the above situation.

(5 marks)